# Alexander Ryabchenko

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## Profile

PhD student at the University of Toronto and Vector Institute researching the statistical and computational foundations of modern machine learning, including post-training, reinforcement learning, and optimization.

## Education

**Doctor of Philosophy (PhD), Statistical Sciences** 2026 – Present
*University of Toronto* *Toronto, ON*
- Advisors: Murat A. Erdogdu, Wenlong Mou, Stanislav Volgushev.

**Honours Bachelor of Science (HBSc), Mathematics and Computer Science** 2021 – 2025
*University of Toronto* *Toronto, ON*
- Cumulative GPA: 4.00/4.00 (Average grade: 98.0%).
- Governor General's Silver Medal (highest academic standing), John Black Aird Scholarship (top student).

## Publications

- **A.R.**, Jian Qian, Wenlong Mou. Adaptively Incorporating Directional Hints into Zeroth-Order Optimization. To appear in Advances in Neural Information Processing Systems (NeurIPS 2026). arXiv:2609.08277.
- **A.R.**, Idan Attias, Daniel M. Roy. Capacity-Constrained Online Convex Optimization with Delayed Feedback. To appear in NeurIPS 2026. arXiv:2606.11711.
- **A.R.**, Idan Attias, Daniel M. Roy. A Reduction from Delayed to Immediate Feedback for Online Convex Optimization with Improved Guarantees. To appear in NeurIPS 2026. arXiv:2602.02634.
- **A.R.**, Wenlong Mou. Reinforcement Learning with Action-Triggered Observations. In the *Proceedings of the 43rd International Conference on Machine Learning* (ICML 2026). arXiv:2510.02149.
- **A.R.**, Idan Attias, Daniel M. Roy. Capacity-Constrained Online Learning with Delays: Scheduling Frameworks and Regret Trade-offs. In the *Proceedings of the 38th Conference on Learning Theory* (COLT 2025), pp. 4973-5014. Oral presentation. arXiv:2503.19856.

## Preprints

- Idan Attias[*], Orin Levy[*], **A.R.**[*], Yishay Mansour, Uri Stemmer. Vanilla Policy Optimization Is Both Optimal and Differentially Private for Stochastic Contextual Bandits. arXiv:2609.33888. Under review.
- Emmanuel Esposito[*], Andrew Jacobsen[*], Hao Qiu[*], **A.R.**[*], Mengxiao Zhang[*]. Optimal Learning in Games with Arbitrary Delays: Best-of-Both-Worlds for Self-Play and Adversarial Opponents. 2026. Under review.

[*]Equal contribution; alphabetical order.

## Awards

**Connaught International Doctoral Scholarship** (50,000 CAD) 2026 – 2031

**Governor General's Silver Medal** 2025
- Canada's most prestigious academic award, established in 1873; awarded to the undergraduate student with the highest academic standing upon graduation.

**John Black Aird Scholarship** 2025
- Award recognizing the top graduating student at the University of Toronto.

**E. A. Robinson Medal in Sciences** 2025
- Medal recognizing the top graduating science student at the University of Toronto.

**Vector Institute Research Grants** (30,000 CAD total; two-time recipient) 2023 – 2024

**University of Toronto International Scholar Award** (180,000 CAD) 2021 – 2025

**Undergraduate Scholarships & Research Awards at the University of Toronto** 2021 – 2025
- MCS Research Award (7,500 CAD)
- Joseph Alfred Whealy In-Course Scholarship (6,000 CAD total; four-time recipient)
- Peter Fantham Memorial Scholarship in Mathematics (2,000 CAD)
- Doris Geddes Scholarship in Mathematics (1,000 CAD)

## Research Experience

**Graduate Student Researcher** 2026 – Present
*Vector Institute* *Toronto, ON*

- Working on the theoretical foundations of modern AI: LLM post-training, scaling laws, optimization.
- Previously completed undergraduate Vector Institute Research Internships (2023, 2024).

**Research Assistant** 2022 – 2026
*University of Toronto, Department of Statistical Sciences* *Toronto, ON*

- Studied the mathematical foundations of modern machine learning, with projects spanning feature learning in neural networks and high-dimensional statistical estimation.
- Completed MCS Summer Research Internship (2022).

**Visiting Researcher** Spring 2023
*Sunnybrook Health Sciences Centre* *Toronto, ON*

- Improved Swin UNETR for 3D segmentation in cancer and musculoskeletal research using dilated attention and a custom loss, outperforming the standard architecture in ablation studies.
- Released the resulting implementation as an open-source codebase and presented the results at the Sunnybrook Orthopaedic Biomechanics Laboratory seminar.

## Technical Skills

**Programming:** Python, C/C++, Bash, Julia.
**Machine Learning:** PyTorch, JAX, Hugging Face Transformers.
**Development Tools:** Git, Linux, Slurm, Docker.

## Workshops & Talks

**A Reduction from Delayed to Immediate Feedback for Online Convex Optimization**. Poster.

- *Learning Theory Summer School & Workshop* (LTSS 2026), Copenhagen, Denmark (Jun 2026).

**Reinforcement Learning with Action-Triggered Observations**. Seminar Talk.

- *UofT Statistical Sciences Research Program* (UTSSRP 2026), Toronto, Canada (Jun 2026).

**Capacity-Constrained Online Learning with Delays**. Poster.

- *Vector AI/ML "Remarkable 2026" Conference*, Toronto, Canada (Feb 2026).
- *Southern Ontario Learning Theory Workshop* (SOLT 2025), Waterloo, Canada (Nov 2025).

## Academic Service & Teaching

**Conference Reviewer** 2026 – Present

- International Conference on Learning Representations (**ICLR**): 2027.
- Neural Information Processing Systems (**NeurIPS**): 2026 (**Top Reviewer**).
- Conference on Learning Theory (**COLT**): 2026.

**Teaching Assistant** Winter 2025
*University of Toronto* *Toronto, ON*

- CSC311, Introduction to Machine Learning.
- CSC363, Computational Complexity and Computability.